

DD_SIP-03 - Discount Assignment - Developer Implementation Guide

Version: 1.0 Status: Draft for review Bundle: 09_catalogs Audience: mid-level backend developers, billing integration developers, frontend developers, QA Parent DD: DD_SIP-03 - Discount Assignment

1. How To Read This Guide

Discount assignment is the operational layer between the discount catalog and billing runtime.

PLM-CFG-04 says what the discount is. SIP-03 says who gets it and when. DIS-OP-01 later decides how the active assignment is applied to a charge/invoice/wallet event.

Do not put invoice amount calculation in SIP-03.

Do not hardcode approval steps in SIP-03. Configurable approvals are owned by EM-CFG-04. SIP-03 calls EM-CFG-04 to evaluate/start approval, then waits for an approval outcome callback. Camunda is involved only when the selected EM-CFG-04 policy uses `approvalMode = BPMN`.

2. Step 1 - Validate Discount Catalog Item

When a user creates an assignment, first load the discount:

```
GET /api/discounts/RETENTION_10PCT_3M
```

Expected:

```
{
  "discountId": "disc-1001",
  "discountCode": "RETENTION_10PCT_3M",
  "effectiveStatus": "ACTIVE",
  "flavor": "PRICE_REBATE",
  "type": "PERCENTAGE",
  "value": 10,
  "eligiblePackageRefs": ["pkg-fiber-100m", "pkg-fiber-200m"],
  "eligibleBillableEventRefs": ["RECURRING_MRC"],
  "startDate": "2026-01-01",
  "endDate": "2026-12-31"
}
```

Reject assignment when:

- discount does not exist;
- discount is not ACTIVE;
- requested validity does not fit policy;
- discount flavor or billable event does not match intended usage.

3. Step 2 - Preview Assignment

Backoffice or sales app calls:

```
POST /api/discount-assignments/preview
Content-Type: application/json

{
  "operatorCode": "WIK",
  "discountCode": "RETENTION_10PCT_3M",
  "scopeType": "SUBSCRIPTION",
  "scopeRefId": "SUB-700045",
  "customerId": "CUS-100045",
  "subscriptionId": "SUB-700045",
  "validFrom": "2026-06-01",
  "validTo": "2026-08-31",
  "sourceChannel": "BACKOFFICE",
  "reasonCode": "RETENTION_SAVE"
}
```

SIP-03 calls owner APIs:

```
GET /api/subscriptions/SUB-700045
GET /api/packages/pkg-fiber-100m
GET /api/franchises/fr-019
```

Preview response:

```
{
  "eligible": true,
  "approvalRequired": false,
  "approvalPolicyCode": null,
  "approvalMode": "AUTO",
  "duplicateFound": false,
  "stackingConflicts": [],
  "runtimeHints": {
    "eligibleBillableEventRefs": ["RECURRING_MRC"],
    "stackingGroupCode": "RECURRING_MRC",
    "priority": 100
  }
}
```

Frontend should show this before creation.

Approval decision comes from EM-CFG-04:

```
POST /api/approvals/evaluate
Content-Type: application/json

{
  "operatorCode": "WIK",
  "moduleCode": "SIP-03",
  "actionCode": "DISCOUNT_ASSIGNMENT_CREATE",
  "subjectType": "DISCOUNT_ASSIGNMENT",
  "context": {
    "discountCode": "RETENTION_10PCT_3M",
    "reasonCode": "RETENTION_SAVE",
    "sourceChannel": "BACKOFFICE",
    "scopeType": "SUBSCRIPTION",
    "durationDays": 92,
    "estimatedValue": 1500.00,
    "currencyCode": "KES",
    "requesterUserId": "usr-care-001"
  }
}
```

Possible EM-CFG-04 response:

```
{
  "approvalRequired": true,
  "policyId": "apol-da-retention-wik-v1",
  "policyCode": "DISCOUNT_ASSIGNMENT_RETENTION",
  "approvalMode": "BPMN",
  "camundaProcessKey": "approval_generic_threshold_v1",
  "firstQueueCode": "CARE_SUPERVISOR"
}
```

4. Step 3 - Create Assignment

```
POST /api/discount-assignments
Idempotency-Key: da-retention-sub-700045-20260601
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "discountCode": "RETENTION_10PCT_3M",
  "scopeType": "SUBSCRIPTION",
  "scopeRefId": "SUB-700045",
  "customerId": "CUS-100045",
  "accountId": "ACC-900045",
  "subscriptionId": "SUB-700045",
  "reasonCode": "RETENTION_SAVE",
  "sourceChannel": "BACKOFFICE",
  "validFrom": "2026-06-01",
  "validTo": "2026-08-31",
  "priority": 100,
  "stackingGroupCode": "RECURRING_MRC"
}
```

Insert:

```

insert into discount_assignment (
  assignment_id,
  operator_code,
  discount_code,
  discount_id,
  scope_type,
  scope_ref_id,
  customer_id,
  account_id,
  subscription_id,
  package_ref,
  campaign_id,
  franchise_id,
  reason_code,
  source_channel,
  valid_from,
  valid_to,
  status,
  priority,
  stacking_group_code,
  metadata_json,
  created_by_user_id,
  created_at,
  activated_at
) values (
  'da-001',
  'WIK',
  'RETENTION_10PCT_3M',
  'disc-1001',
  'SUBSCRIPTION',
  'SUB-700045',
  'CUS-100045',
  'ACC-900045',
  'SUB-700045',
  'pkg-fiber-100m',
  null,
  null,
  'RETENTION_SAVE',
  'BACKOFFICE',
  '2026-06-01',
  '2026-08-31',
  'ACTIVE',
  100,
  'RECURRING_MRC',
  '{"previewId":"dap-001"}',
  'usr-care-001',
  now(),
  now()
);

```

Add history:

```

insert into discount_assignment_status_history (
  history_id,
  assignment_id,
  old_status,
  new_status,
  reason_code,
  changed_by_user_id,
  changed_at
) values (
  'dash-001',
  'da-001',
  null,
  'ACTIVE',
  'CREATED_AUTO_ACTIVE',
  'usr-care-001',
  now()
);

```

Emit DiscountAssignmentActivated.

5. Step 4 - Approval Required Flow

If EM-CFG-04 says approval is required, create the assignment as PENDING_APPROVAL, then start an EM-CFG-04 approval request.

```

insert into discount_assignment_approval (
  approval_id,
  assignment_id,

```

```

    approval_policy_code,
    approval_request_id,
    approval_mode,
    camunda_process_instance_id,
    status,
    assigned_queue_code,
    created_at
) values (
    'daa-001',
    'da-002',
    'DISCOUNT_ASSIGNMENT_RETENTION',
    null,
    'BPMN',
    null,
    'PENDING',
    'CARE_SUPERVISOR',
    now()
);

```

Start approval in EM-CFG-04:

```

POST /api/approvals/requests
Idempotency-Key: approval-da-002
Content-Type: application/json

```

```

{
  "operatorCode": "WIK",
  "moduleCode": "SIP-03",
  "actionCode": "DISCOUNT_ASSIGNMENT_CREATE",
  "subjectType": "DISCOUNT_ASSIGNMENT",
  "subjectId": "da-002",
  "policyId": "apol-da-retention-wik-v1",
  "callbackUrl": "/internal/discount-assignments/da-002/approval-outcome",
  "context": {
    "discountCode": "RETENTION_20PCT_3M",
    "reasonCode": "RETENTION_SAVE",
    "scopeType": "SUBSCRIPTION",
    "scopeRefId": "SUB-700045",
    "durationDays": 92,
    "estimatedValue": 3000.00,
    "currencyCode": "KES"
  }
}

```

Expected response:

```

{
  "approvalRequestId": "apr-da-002",
  "status": "PENDING",
  "approvalMode": "BPMN",
  "camundaProcessInstanceId": "cam-proc-da-002",
  "firstTaskId": "apt-da-002-step1"
}

```

Patch local reference:

```

update discount_assignment_approval
set approval_request_id = 'apr-da-002',
    camunda_process_instance_id = 'cam-proc-da-002'
where approval_id = 'daa-001';

```

Approver approves in the shared approval UI:

```

POST /api/approvals/tasks/apt-da-002-step1/approve
Content-Type: application/json

{
  "decisionComment": "Approved 20% retention discount for customer save."
}

```

EM-CFG-04 finalizes the approval and calls SIP-03:

```

POST /internal/discount-assignments/da-002/approval-outcome
Content-Type: application/json

{
  "approvalRequestId": "apr-da-002",
  "outcome": "APPROVED",
  "decisionComment": "Approved 20% retention discount for customer save.",
  "decidedByUserId": "usr-supervisor-002"
}

```

SIP-03 applies the outcome idempotently:

```

update discount_assignment_approval

```

```

set status = 'APPROVED',
  actor_user_id = 'usr-supervisor-002',
  decision_comment = 'Approved 20% retention discount for customer save.',
  decided_at = now()
where assignment_id = 'da-002'
  and approval_request_id = 'apr-da-002'
  and status = 'PENDING';

update discount_assignment
set status = 'ACTIVE',
  activated_at = now()
where assignment_id = 'da-002'
  and status = 'PENDING_APPROVAL';

```

If the outcome is REJECTED, set `discount_assignment.status = REJECTED` and keep it excluded from runtime queries.

6. Step 5 - Runtime Query

DIS-OP-01 calls SIP-03 when rating/charging an event:

```
GET /api/discount-assignments/effective?operatorCode=WIK&subscriptionId=SUB-700045&eventCode=RECURRING_MRC&onDate=2026-06-30
```

SIP-03 query logic:

```

select *
from discount_assignment
where operator_code = 'WIK'
  and status = 'ACTIVE'
  and valid_from <= date '2026-06-30'
  and (valid_to is null or valid_to >= date '2026-06-30')
  and (
    (scope_type = 'SUBSCRIPTION' and scope_ref_id = 'SUB-700045')
    or (subscription_id = 'SUB-700045')
  )
order by priority desc, created_at asc;

```

SIP-03 returns assignments only. DIS-OP-01 loads discount catalog details and computes actual application.

7. Step 6 - Cancel Assignment

Cancel future use:

```

POST /api/discount-assignments/da-001/cancel
Content-Type: application/json

{
  "reasonCode": "CUSTOMER_NO_LONGER_ELIGIBLE",
  "effectiveDate": "2026-07-01",
  "comment": "Customer changed package."
}

```

Patch:

```

update discount_assignment
set status = 'CANCELLED',
  cancelled_at = now(),
  metadata_json = metadata_json || '{"cancelReasonCode":"CUSTOMER_NO_LONGER_ELIGIBLE","cancelEffectiveDate":"2026-07-01"}'
where assignment_id = 'da-001'
  and status in ('ACTIVE', 'SUSPENDED', 'PENDING_APPROVAL');

```

Important: do not reverse already issued invoices. If a past invoice must be corrected, create a BIL-02-ADJ-01 adjustment request.

8. Step 7 - Bulk Assignment

Commercial user creates a campaign batch:

```

POST /api/discount-assignment-batches
Content-Type: application/json

{
  "operatorCode": "WIK",
  "batchType": "CAMPAIGN",
  "discountCode": "ACQ_FREE_INSTALL",

```

```

    "campaignId": "camp-acq-june",
    "validFrom": "2026-06-01",
    "validTo": "2026-06-30",
    "targets": [
      {"targetRefType": "ORDER", "targetRefId": "ORD-1001"},
      {"targetRefType": "ORDER", "targetRefId": "ORD-1002"}
    ]
  }
}

```

Create batch:

```

insert into discount_assignment_batch (
  batch_id,
  operator_code,
  batch_type,
  discount_code,
  campaign_id,
  requested_by_user_id,
  status,
  total_rows,
  valid_rows,
  failed_rows,
  created_at
) values (
  'dab-001',
  'WIK',
  'CAMPAIGN',
  'ACQ_FREE_INSTALL',
  'camp-acq-june',
  'usr-commercial-001',
  'VALIDATING',
  2,
  0,
  0,
  now()
);

```

Worker validates each row, then creates assignments using idempotency key:

```
discount-batch:dab-001:ORDER:ORD-1001
```

Successful item:

```

insert into discount_assignment_batch_item (
  batch_item_id,
  batch_id,
  target_ref_type,
  target_ref_id,
  status,
  assignment_id
) values (
  'dabi-001',
  'dab-001',
  'ORDER',
  'ORD-1001',
  'CREATED',
  'da-101'
);

```

9. Suggested Worker Commands

These can run inside the existing catalog/commercial worker pool.

```

discount-assignment:expire
discount-assignment:batch-run
discount-assignment:approval-reminder
discount-assignment:runtime-cache-refresh

```

Command	Purpose
discount-assignment:expire	Marks assignments EXPIRED after valid_to.
discount-assignment:batch-run	Validates and processes batch items.
discount-assignment:approval-reminder	Alerts approval queues about pending approvals.
discount-assignment:runtime-cache-refresh	Refreshes active assignment read model/cache for DIS-OP.

10. Drools Rule Samples

Use Drools for configurable assignment eligibility, stacking, and approval decisions.

```
package rules.commercial.discount_assignment

rule "Target package must be eligible for discount"
when
    $a : DiscountAssignmentFact(packageEligible == false)
    $out : DiscountAssignmentDecision()
then
    $out.reject("DA_TARGET_NOT_ELIGIBLE");
end

rule "Same stacking group conflict blocks assignment"
when
    $a : DiscountAssignmentFact(hasActiveStackingConflict == true)
    $out : DiscountAssignmentDecision()
then
    $out.reject("DA_STACKING_CONFLICT");
end

rule "Long manual retention assignment requires approval"
when
    $a : DiscountAssignmentFact(sourceChannel == "BACKOFFICE", reasonCode == "RETENTION_SAVE", durationDays > 90)
    $out : DiscountAssignmentDecision()
then
    $out.setApprovalRequired(true);
    $out.setApprovalPolicyCode("DISCOUNT_ASSIGNMENT_RETENTION");
end
```

11. Error Codes

Code	Meaning
DA_DISCOUNT_NOT_FOUND	Discount catalog item does not exist.
DA_DISCOUNT_NOT_ACTIVE	Discount is inactive/retired.
DA_TARGET_NOT_FOUND	Scope target does not exist.
DA_TARGET_NOT_ELIGIBLE	Target violates discount eligibility.
DA_DUPLICATE_ACTIVE_ASSIGNMENT	Duplicate active assignment blocked.
DA_STACKING_CONFLICT	Existing assignment conflicts by stacking group.
DA_APPROVAL_REQUIRED	Assignment cannot activate before approval.
DA_APPROVAL_REQUEST_FAILED	EM-CFG-04 approval request could not be started.
DA_APPROVAL_OUTCOME_MISMATCH	Approval callback does not match assignment/policy/request.
DA_BATCH_PARTIALLY_FAILED	Bulk batch has failed rows.
DA_CANNOT_CANCEL_TERMINAL_STATUS	Assignment is already expired/cancelled/rejected.

12. Minimum Implementation Order

1. Create assignment tables and indexes.
2. Implement discount catalog lookup adapter.
3. Implement target lookup adapters for customer/account/subscription/order/franchise.
4. Implement preview API.
5. Implement create assignment API with idempotency.
6. Implement EM-CFG-04 approval evaluate/request/callback integration.
7. Implement cancel API.
8. Implement runtime effective assignment query.
9. Implement batch tables and batch worker.
10. Emit assignment events.

11. Patch Backoffice assignment screens.
12. Patch Sales App eligibility display/request.
13. Add E2E tests.

13. Test Scenarios

13.1 Manual Retention Discount

- Care user previews subscription assignment.
- Target is eligible.
- Assignment is created active.
- Runtime query returns assignment.

13.2 Approval Required

- User creates high-value discount.
- Status is PENDING_APPROVAL.
- Runtime query excludes it.
- Supervisor approves through EM-CFG-04 task API.
- EM-CFG-04 callback activates assignment.
- Runtime query includes it.

13.3 Duplicate Blocked

- Active assignment exists for same discount/scope/window.
- New create returns 409 DA_DUPLICATE_ACTIVE_ASSIGNMENT.

13.4 Campaign Batch

- Batch has 100 targets.
- 96 assignments created.
- 4 failed rows stored with failure code.
- Batch status becomes PARTIALLY_FAILED.

13.5 Cancellation

- Assignment is cancelled.
- Runtime query no longer returns it for future date.
- Past invoice remains unchanged.

14. Developer Checklist

- Assignment validates against PLM-CFG-04.
- Assignment target is validated through owning module API.
- Runtime query returns assignments, not calculated amounts.
- Approval is configured and executed by EM-CFG-04; SIP-03 stores only the local approval reference and final outcome.
- Camunda is used only when EM-CFG-04 returns approvalMode = BPMN.
- Drools covers configurable assignment eligibility/stacking/approval decisions.
- Cancellation does not create invoice/wallet reversal.
- Bulk processing is idempotent per target.

- Events support DIS-OP cache refresh and reporting.